

Notification Distribution — IT Build Handoff

For: IT / Power Platform team **From:** Dan Plant, Technical Superintendent, Well Control Engineering **Date:** 2026-08-27

Companion file: Seadrill_WCE_Notification_Distribution_Matrix.xlsx (maintained by Dan — the source of truth for WHO gets WHAT)

1. What this is and why

The WCE reporting pipeline already delivers every offshore report to a SharePoint folder automatically. What it does NOT do is tell the right people: today the only notification recipients anywhere in the system are Dan Plant and Lee (Head of WCE), hard-wired.

This handoff asks IT to build the notification layer: when a report file arrives, work out **which vessel, which report type**, and **how severe** it is, then email the people the distribution matrix names for that combination. The matrix is standardised **by position** (one rule set for the whole fleet); a roster sheet maps position → name/email **per vessel**.

The existing pipeline (do not change any of it)

```
Offshore HTML report tools
  | HTTP POST {FileName, ContentType, FileContent(base64)}
  v
Power Automate flow (existing - "posting flow")
  | writes the file
  v
SharePoint: WellControl / PostedReports          <== YOUR TRIGGER POINT
  | OneDrive sync
  v
Update-Dashboard.ps1 (scheduled on Dan's PC) -> static dashboards on IIS
```

Hard rule from Dan: the posting flow and the file transport must not be modified. Build the notification logic as a **separate, new flow** triggered by file creation in the PostedReports folder ("When a file is created (properties only)" trigger, or equivalent). If the new flow breaks, files must still land and dashboards must still build.

2. The source of truth: the distribution matrix workbook

Seadrill_WCE_Notification_Distribution_Matrix.xlsx, four tabs:

Tab	Contents
README	Instructions + legend (for humans, not the flow)
Escalation Levels	Definitions of L1/L2/L3 and the data signals behind them
Distribution Matrix	Report type × escalation level rows; 15 position columns; each cell is TO, CC, or blank
Vessel Roster	Vessel × Position → Name, Email

Distribution Matrix tab layout

- Header row: row 4. Data rows: 5–26 (22 rows).
- Columns: A = Report type, B = Level (L1/L2/L3), C = Trigger description (human text), **D–R = the 15 positions**, S = TO-count formula (ignore).
- Positions D–R, in order: OIM, Subsea Supervisor, Subsea Engineer, Rig Engineer, Chief Engineer, Electrical Supervisor, Maintenance Supervisor, WCE Technical Superintendent, Head

of WCE, Subsea Superintendent, Rig Manager, Technical Section Lead, Operations Manager, Marine Superintendent, Planner.

- Cell values: **TO** = direct recipient, **CC** = copied, blank = not notified.

Vessel Roster tab layout

- Columns: A = Vessel, B = Position, C = Name, D = Email.
- The first block is vessel **ALL (Fleet)** — fleet-wide onshore roles entered once (WCE Technical Superintendent, Head of WCE, etc.).
- Then one block per vessel with the offshore positions plus per-vessel Rig Manager and Technical Section Lead.
- **Precedence:** a per-vessel row overrides an ALL (Fleet) row for the same position on that vessel.
- **Blank email = position not carried / not yet filled → silently skip that position for that vessel.** This is normal, not an error. (E.g. Rig Engineer exists only on West Auriga, West Jupiter, West Tellus and West Polaris — only those four vessels have Rig Engineer roster rows.)
- Row 5 is a light-blue EXAMPLE row — ignore rows whose Name contains “EXAMPLE”.

Recommended runtime storage

Reading the .xlsx directly from a flow is fragile (table addressing, locks while Dan edits).

Recommended: mirror the two data tabs into two **SharePoint lists** the flow reads:

- **WCE_DistributionMatrix:** ReportType, Level, Position, Role (TO/CC) — one row per non-blank matrix cell (~150 rows).
- **WCE_VesselRoster:** Vessel, Position, Name, Email.

The workbook remains the master that Dan maintains; agree a simple re-sync path (a small flow that re-imports on workbook change, or a manual re-import on request — Dan updates rarely). If your team prefers the Excel Online (Business) connector against named tables in the workbook, that is acceptable too — convert the ranges to Excel tables first and move the workbook into SharePoint.

3. Classifying an incoming file

Everything lands in PostedReports as a file. Classification uses the filename first, then fields inside the JSON.

3a. Report type

Filename / content test (in this order)	Report type (matrix row key)
Filename starts ssce-request_	SSCE / COC Request Submitted
Filename starts ssce-decision_	SSCE / COC Request Decision
Filename starts seadrill-daily-checks_	Daily Checks (Rig Monitoring) — meta.reporttype distinguishes Daily Checks vs FLM; FLM files carry the FLM report type
JSON meta.discipline = “Marine”	Marine Integrity Report
JSON meta.reporttype present	Use its value: Daily Log → Daily Log / Lessons Learned; Precharge-type → Precharge Report; etc.
A tile contains planningData	BOP Planning Report
A tile contains bwmData	BWM Weekly Planning (Excel)
A tile contains r53Data	S53 Failure Report (SSORT)

Filename / content test (in this order)	Report type (matrix row key)
A tile contains cbmData	CBM Inspection Report
None of the above	Rig Visit Report

The same routing already exists in `scripts/Update-Dashboard.ps1`, function `Get-ReportType` — treat that as the reference implementation. The `ssce-request_ / ssce-decision_` filename prefixes are a fixed contract and will never change.

3b. Vessel

- Most reports: `JSON meta.asset`.
- Daily Checks / FLM: `meta.checks.rig` (fallback `meta.asset`).
- SSCE requests: `applicant.siteUnit`.
- Match against the Vessel column of the roster case-insensitively, ignoring hyphen/space differences (West-Saturn = West Saturn).

3c. Escalation level

Default is **L1**. Elevate when the report's own data says so:

Report type	L2 signal	L3 signal
Daily Checks / FLM	Any reading in <code>meta.checks.values</code> whose value is “fail” or whose <code><key>_cmt</code> companion is non-empty (comment presence is the real attention signal)	—
S53 Failure Report	<code>r53Data.fields.s53_isfailure</code> indicates a failure	Rig on downtime / NPT recorded — field name to be confirmed with the reporting-tool owner (see §7); until confirmed, treat all S53 failures as L2
CBM Inspection	Any component graded poor	—
Marine Integrity	Any <code>mi_pol_*</code> / <code>mi_reg_*</code> / <code>mi_eqp_*</code> key with score 1 or 2 (keys ending <code>_cmt</code> are comments, not scores)	—
Precharge	Reading out of tolerance	—
BOP Planning	Slippage / red status flagged	—
Break-in work	Always at least L2 (unplanned by nature)	—
SSCE Request	—	<code>applicant.priorityLevel</code> = “1-High” and <code>applicant.rigOnDowntime</code> = “Yes”

Missing-row rule: if the computed level has no row in the matrix for that report type, use the nearest lower level that does exist (L3→L2→L1).

4. Recipient lookup (per incoming file)

```

type = classify(file)           # 3a
vessel = extract(file)         # 3b
level = severity(file)         # 3c, with missing-row fallback

rows = Matrix[ReportType = type AND Level = level]
for each position in rows:
    entry = Roster[Vessel = vessel, Position = position] # per-vessel first
    if none: entry = Roster[Vessel = "ALL (Fleet)", Position = position]
    if entry exists and Email non-blank:
        add email to TO or CC per the matrix cell
if TO list is empty: fall back to section 5 defaults
send one email per report file

```

Email content

- Subject: [WCE <L1|L2|L3>] <Report type> – <Vessel> – <date> (prefix L2 with △ and L3 with a red marker if the mail client renders it).
- Body: vessel, report type, level and its trigger text (matrix column C), plus a **deep link straight to the report** — the dashboard supports it:
<http://sdrllazneuiis01d.corp.local:8080/sacred/dashboard/dashboard.html?report=<FileName>>
- One email per report, not per recipient. De-duplicate a person who appears via two positions (TO wins over CC).

5. Edge rules and fallbacks

- **Unknown report type or unrecognisable vessel:** notify Dan Plant + Head of WCE only (the current behaviour), subject prefixed [WCE UNROUTED], so nothing ever disappears silently.
- **Roster gaps** (position marked TO/CC but no email for that vessel): skip silently — see §2. If the resulting TO list is empty, apply the unrouted fallback above.
- **Duplicate/re-posted files:** the pipeline is newest-file-wins; the same logical report can be re-posted with a new filename. Acceptable for v1 to re-notify on re-post. If you want to suppress, key on vessel+type+report date.
- **Pilot / parallel run:** for the first 2-4 weeks, ALWAYS add Dan + Lee as CC on every send regardless of the matrix, so the current guaranteed audience is preserved while the matrix is validated. Remove once Dan signs off.

6. What already exists — read before building, to avoid double-sends

1. **The scanner has a basic notifier** (*Update-Dashboard.ps1*, config-driven: a notifications block in config.json, optional notification-rules.csv of ReportType→Email, sends via local Outlook or SMTP relay from Dan's PC). It is type-only — no vessel, no severity, no TO/CC. Once the Power Automate flow is live it should be left disabled so people don't get two emails. Its rules format is NOT the matrix; ignore it for this build.
2. **Two event feeds already exist for special cases** — the scanner writes them for exactly this purpose (“files for an external flow to pick up”): *break-ins-pending.json* (unplanned work items) and *ssce-notifications-pending.json* (SSCE submitted / decided events with requestId, rig, priority). If it is easier to drive the Break-in and SSCE-decision notifications from these files than from raw report files, that is a supported option — dedup state is already handled scanner-side.
3. **BWM Weekly Planning is an Excel snapshot, not a posted JSON** — if notifications are wanted for it, the trigger is “file modified” on the planning Excel path, not PostedReports. Fine to descope from v1; the matrix row exists for when it's wanted.

7. Open items

#	Item	Owner
1	Confirm the S53 downtime / NPT field name in the SSORT export so the S53 L3 trigger can be automated (until then all S53 failures notify at L2)	Dan → reporting-tool owner
2	Decide runtime storage: SharePoint lists (recommended) vs Excel Online tables	IT
3	Agree the workbook→list re-sync mechanism and who runs it	IT + Dan
4	Sending identity: shared mailbox (e.g. wce-dashboard@seadrill.com) recommended over a personal account	IT
5	Vessel roster names/emails — being filled in by Dan now	Dan

8. Acceptance test checklist

Run with test addresses before pointing at the real roster:

1. Post a clean Daily Checks file (all pass) → L1 recipients only, for the right vessel.
2. Post a Daily Checks file containing one “fail” value → L2 row fires; L1-only recipients not duplicated; △ subject.
3. Post an ssce-request_* file with priorityLevel “1-High” and rigOnDowntime “Yes” → L3 row fires.
4. Post a report for a vessel with a blank roster email in one TO position → that person skipped, everyone else notified, no error.
5. Post a Rig Engineer-relevant L2 report for West Saturn (no Rig Engineer) and for West Auriga (has one) → column skipped on Saturn, honoured on Auriga.
6. Post a file with an unknown type → [WCE UNROUTED] to Dan + Lee only.
7. Confirm the deep link in the email opens the exact report on the dashboard.
8. Confirm the existing posting flow and dashboard build are untouched throughout.

Questions on the matrix content → Dan Plant. Questions on the pipeline internals → INTEGRATION-CONTRACT.md in the project repo is the full technical contract; Update-Dashboard.ps1 is the reference implementation for all classification rules.